

Business Analysis and Valuation

Group Project

Taiwan Semiconductor Manufacturing Company

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Executive Summary

TSMC is the world's largest contract chip manufacturer. It makes the chips that Apple, NVIDIA, and AMD design but don't fabricate themselves, and it has roughly 70% of the global foundry market and over 90% of leading-edge production. We were asked to recommend whether to buy the stock, and our answer is HOLD.

TSMC closed at 2,185 TWD on April 24, 2026, basically at its all-time high. Across our four valuation methods the per-share value range was 1,545 to 2,367 TWD with a median of 1,905. That puts the stock about 13% above where we think it should trade. Not enough to call a SELL, but definitely not a BUY at these prices.

What we found

TSMC's accounting is unusually clean. Equipment is depreciated over five years (which is conservative), R&D is fully expensed (also conservative), goodwill is 0.07% of assets, and stock-based comp is 0.04% of revenue. There are three areas where management discretion creates room for the numbers to move (timing of depreciation, unrecognized DTL on foreign earnings, returns reserve sizing), but none rises to the level of a traditional red flag.

The Abnormal Earnings model gave us 1,545 TWD per share, about 29% below market. We used a 41% net margin in the base case, which is the three-year average. That looks low next to Q1 2026's 50.5% margin, but we wanted a deliberately conservative floor.

The most striking finding came from the ROE mean reversion analysis. Even if we assume TSMC holds its current 35% ROE for the next decade, the implied value is only 1,886 TWD. The market is pricing in something better than 'sustain forever' and we struggle to justify that at face value.

The gap between our value and the market price is mostly about expectations, not risk. The stock price implies long-run growth of about 4.4%, well above our 2.5% terminal assumption. That's a big assumption to make about a company TSMC's size.

1. Industry and Strategic Analysis

1.1 Industry

TSMC manufactures chips that other companies design. It does not sell its own branded chips, which is what makes it a foundry rather than an integrated chip company like Samsung or Intel. The market is an oligopoly: TSMC, Samsung Foundry, Intel Foundry, GlobalFoundries, and UMC. TSMC dominates the leading edge (3nm and below) with over 90% share, and the trailing-edge market is more competitive.

Capital intensity is what makes this industry interesting. TSMC spent 1.27 trillion TWD (about 40 billion USD) on capex in 2025, equal to 33% of revenue. A new leading-edge fab costs 20 to 30 billion USD and takes three to four years to build. Those numbers keep new entrants out.

1.2 Five Forces

Threat of new entry is essentially zero at the leading edge. The capital and technology hurdles have only grown over time, and even Intel has struggled to catch up despite massive government subsidies.

Buyer power is lower than you'd expect given the customer concentration. Apple and NVIDIA together represent more than a third of revenue, but switching foundries at the leading edge is a multi-year project. Samsung Foundry has tried for years to win Apple back; it hasn't worked.

Supplier power is real in one specific place: ASML's EUV lithography tools, which have no substitute. Everywhere else, TSMC has multiple sources for chemicals, gases, and equipment.

Substitutes and rivalry are both manageable. There is no real substitute for advanced silicon at this stage of the technology cycle. Rivalry exists but it's structurally muted by the capacity constraints we just described.

Putting it together: TSMC operates in an industry where it has substantial pricing power, locked-in customers, and very high returns on capital. The structure is unusually favorable.

Suppliers (249) »				Customers (136) »			
Name	Supplier's Rev %	TSM's Cost % ↓	3M Price Chg. ↗	Name	TSM's Rev % ↓	Customer's Cost %	3M Price Chg. ↗
ASML Holding NV	16.65%	15.23% CAPEX	18.45%	NVIDIA Corp	19.04%	37.35% COGS	6.28%
Applied Materials Inc	18.00%	15.08% CAPEX	31.96%	Apple Inc	17.01%	7.70% COGS	-0.48%
Lam Research Corp	18.00%	8.37% CAPEX	30.65%	Advanced Micro Devices Inc	5.20%	37.35% COGS	13.23%
Tokyo Electron Ltd	11.54%	5.41% CAPEX	2.61%	QUALCOMM Inc	5.11%	30.96% COGS	-19.89%
KLA Corp	17.00%	4.88% CAPEX	25.16%	Broadcom Inc	4.23%	22.95% COGS	11.85%
Advantest Corp	12.33%	1.85% CAPEX	20.05%	Amazon.com Inc	4.09%	1.25% COGS	5.83%
Teradyne Inc	19.03%	1.79% CAPEX	59.14%	MediaTek Inc	3.83%	49.50% COGS	15.44%
SCREEN Holdings Co Ltd	14.35%	1.73% CAPEX	32.70%	Intel Corp	2.54%	9.82% COGS	28.88%
Daiichi Sankyo Co Ltd	13.80%	1.50% CAPEX	19.64%	Marvell Technology Inc	1.20%	37.35% COGS	64.72%
Shin-Etsu Chemical Co Ltd	4.38%	1.45% COGS	18.39%	Cisco Systems Inc	1.16%	7.24% COGS	10.32%
Lasertec Corp	26.30%	1.18% CAPEX	19.45%	Sony Group Corp	1.13%	2.21% COGS	-15.98%
Tokyo Ohka Kogyo Co Ltd	33.60%	1.08% COGS	49.57%	Infineon Technologies AG	1.02%	13.48% COGS	7.18%
Entegris Inc	16.00%	1.04% COGS	33.94%	STMicroelectronics NV	0.86%	13.47% COGS	44.18%

1.3 Strategy

TSMC's strategy has been remarkably consistent for almost four decades. It commits exclusively to the foundry model, which avoids the conflict of interest that hurt Samsung and Intel when they tried to manufacture chips for customers who were also competitors. It pushes process technology aggressively (N7 in 2018, N5 in 2020, N3 in 2022, N2 in late 2025). It invests in the customer ecosystem through its Open Innovation Platform and its tight partnerships with EDA vendors. And more recently, it has begun expanding geographically into Arizona, Kumamoto, and Dresden as a hedge against Taiwan-specific risks.

1.4 Versus Peers

TSMC's profitability is in a different league from its closest peers. The gap to UMC, MediaTek, and the OSAT players (assembly and test) is wide and getting wider, not narrowing.

I. Depreciation Policies and Asset Distortion

- UMC estimates the useful life of its machinery at 6 years. TSMC's 5-year policy is more conservative, leading to faster expense recognition and lower risk of carrying obsolete equipment at book value compared to UMC. Broadcom on the other hand uses a weighted-average life of 6 years for its technology assets.

II. Revenue Quality and Receivables

- MediaTek allows credit periods of 30 to 90 days, and Broadcom carries a significantly higher trade receivable balance relative to revenue than TSMC. TSMC's shorter cycle of 27 days indicates higher-quality revenue and lower risk of channel stuffing.

III. Grant Accounting and Earnings Smoothing

- TSMC follows an asset reduction where the grant is netted against assets to lower future costs. UMC follows a deferred income approach which records grants as deferred income, recognizing them as other income on a straight-line basis. This makes UMC's operating costs appear higher on the face of the income statement compared to TSMC, while TSMC's method provides a cleaner gross margin.

IV. Inventory Risk Profiles

- Foundry companies like TSMC and UMC carry mostly WIP, while fabless firms such as MediaTek and Broadcom carry significantly more finished goods. Because finished goods are susceptible to consumer market price drops, MediaTek and Broadcom face a higher risk of unnatural inventory write-downs. TSMC's auditor reports significantly fewer concerns regarding slow-moving inventory than those found in UMC's and MediaTek's filings. TSMC's foundry model mitigates risk through pre-contracted WIP with guaranteed buyers. In contrast, fabless peers like MediaTek and mature-node foundries like UMC face higher obsolescence risks due to speculative finished goods and cyclical market exposure.

2. Accounting Analysis

2.1 Reporting Framework

TSMC reports under IFRS as adopted by Taiwan's Financial Supervisory Commission, with a US GAAP reconciliation in its 20-F. Every audit opinion in our period was unqualified. All figures in this report are in millions of TWD unless we say otherwise.

2.2 Revenue Recognition

TSMC recognizes wafer revenue over time as production progresses, under IFRS 15. There are no take-or-pay structures or multi-year contracts that could distort revenue timing.

DSO has actually improved over our period, dropping from 40 days in 2020 to 27 days in 2025. Collections are getting faster, not slower, which is the opposite of what you'd see from a company that was stretching to recognize revenue.

2.3 Property, Plant and Equipment

This is the most consequential accounting choice TSMC makes. Equipment is depreciated over five years on a straight-line basis. Buildings depreciate over 20 to 50 years and land isn't depreciated.

Five years is conservative. The actual economic life of a leading-edge tool is closer to 7 to 10 years, and competitors typically use longer lives. UMC, for example, uses six years for machinery; Broadcom uses a six-year weighted average for technology assets. The reason TSMC's choice makes sense for TSMC specifically is that the foundry business is built around

relentless node transitions. By the time the market has moved from 5nm to 3nm to 2nm, equipment built for the older node has lost most of its leading-edge economic value, regardless of its mechanical condition. Matching the depreciation schedule to the technology cycle (rather than the physical life of the machine) keeps the balance sheet from carrying assets at values the market would never pay.

The result is that TSMC's depreciation expense overstates true economic depreciation, reported earnings are below cash earnings, and the equipment carrying value on the balance sheet is suppressed.

We see this clearly in CFO/NI, which has averaged 1.55x over our period. That's well above the 1.0x to 1.2x you'd expect from a typical capital-intensive business. ROE is also somewhat understated for the same reason: book value carries less depreciation expense than economic reality would justify.

We don't make any adjustment here. The policy is consistent year over year, so cross-period comparisons remain valid. And conservatism in reported earnings is a feature, not a bug, when we're trying to value the company.

2.4 Research and Development

Under IFRS, research has to be expensed but development can be capitalized when it meets certain criteria. TSMC takes the more conservative path and expenses everything as incurred.

R&D as a share of revenue has fallen from 8.2% in 2020 to 6.5% in 2025, but only because revenue grew faster than the R&D budget. Absolute R&D spending has gone up every year. If TSMC capitalized R&D and amortized it over five years, equity would be roughly 13% higher. We're not making that adjustment because we want our analysis to tie to the reported figures, not because there's a problem with the policy.

2.5 Tax Position

Taiwan's statutory corporate tax rate is 20%, but TSMC's effective rate has run well below that for years: 12.9% in 2020, climbing to 17.1% in 2025. The reconciliation involves a few different things. Investment tax credits under Taiwan's Statute for Industrial Innovation reduce the cash tax liability. Hsinchu, Tainan, and Taichung science parks offer partial tax holidays for new fab capacity in the early years. And TSMC has subsidiaries in Singapore and Netherlands that operate at lower rates. There's also 62.9 billion TWD of net deferred tax assets at end-2025.

The trend is what matters. The effective rate stepped up about four points between 2020 and 2025, partly because the OECD Pillar Two minimum tax kicked in. Our forecast uses 20% (statutory) as a deliberately conservative choice; the actual rate will likely come in below that, which builds some cushion into our valuation.

2.6 Stock-Based Compensation

TSMC's SBC was 1,246 million TWD in 2025 against 3,809 billion TWD of revenue, or 0.03%. That's the lowest SBC ratio in the large-cap technology peer set we looked at. NVIDIA runs around 4%, AMD around 5%. There is no SBC-driven earnings inflation to worry about with TSMC.

2.7 Working Capital and Inventory Mix

Working capital is clean and getting cleaner. The cash conversion cycle has compressed from 115 days at the 2021 peak to 76 days at end-2025. DSO went from 40 to 27 days, DIO from 92 to 69 days.

What's worth pausing on is the inventory composition. Roughly 65% of TSMC's inventory is work-in-process versus a much smaller finished goods balance. That mix is what you'd expect from a make-to-order foundry: production starts only after a customer has committed to a contract, so the inventory is effectively pre-sold. This matters for earnings quality. Peers like MediaTek and Broadcom carry much more finished goods, which exposes them to inventory write-downs when end-market prices drop. TSMC has very little of that risk because almost nothing in inventory is sitting unsold.

2.8 Goodwill and Acquisitions

Goodwill at end-2025 was 5,891 million TWD, or 0.07% of total assets. Other intangibles were another 0.24%. For context, AMD and Broadcom both have made multi-billion-dollar acquisitions; TSMC has effectively grown organically. The absence of goodwill is itself a positive signal because it removes a major source of future impairment risk.

2.9 One-Time Items

Restructuring charges over the full 2020 to 2025 window totaled 2,531 million TWD, or 0.02% of cumulative revenue. Asset writedowns peaked at 1,150 million in 2024 (mature-node fab equipment retirement). All of this is small enough that we ignore it.

2.10 Government Grants

TSMC received NT\$76.3 billion in government grants in 2025, primarily tied to its overseas fab build-outs (Arizona under the US CHIPS Act, Kumamoto from the Japanese government, Dresden from the EU). The accounting choice it made is to net these grants against the carrying amount of the related assets rather than recording them as deferred income or other revenue.

What this does in practice is lower the book value of those fab assets, which in turn lowers depreciation expense over the assets' five-year life. The benefit shows up as a slightly lower COGS each year rather than as a one-time gain or as recurring 'other income.' UMC takes the opposite approach and records grants as deferred income recognized on a straight-line basis, which makes UMC's operating costs look higher and its 'other income' line larger relative to TSMC's.

The TSMC method gives a cleaner gross margin number, but it does blur the line between subsidized and unsubsidized cost of production. For a reader trying to figure out the true economics of a fab built without government help, the disclosure isn't quite as transparent as the deferred income approach would be. We don't make any adjustment for this, but we want to flag the choice.

2.12 Net Cash Position

Worth flagging because it changes how we think about the valuation. TSMC has 3,068 billion TWD of cash and short-term investments against 1,068 billion of debt, giving net cash of about 2 trillion TWD (roughly 63 billion USD). That generated 105.7 billion TWD of interest income in 2025, or 5.2% of net income.

There are two implications. TSMC is essentially a creditor rather than a borrower, so the WACC calculation is heavily weighted toward the cost of equity. And the AE valuation includes the cash inside book value, where it earns roughly 3.5% pretax, well below the cost of equity. That technically destroys some abnormal earnings in the model. The DCF handles it more cleanly by adding net cash separately at face value, which is why our DCF result comes in above our AE result.

2.13 Adjustments and Conclusion

The adjustments we make for valuation are minor:

- Use Total Common Equity of 5,355,039 million TWD as the AE base. Minority interest of 41,180 million is excluded.
- Forecast at the 20% statutory tax rate. Conservative versus the historical 13 to 17%.
- No depreciation policy adjustment. Five years is conservative and consistent.
- No R&D capitalization. Full expensing is conservative and we keep it.
- Recognize the net cash explicitly in the DCF. The AE method retains it inside book value.

3. Financial Analysis

3.1 Revenue and Profitability Through Time

Metric	2020	2021	2022	2023	2024	2025
PROFITABILITY						
Revenue (TWD mm)	-	1,587,415	2,263,891	2,161,736	2,894,308	3,809,054
Revenue Growth YoY	-	18.5%	42.6%	(4.5%)	33.9%	31.6%
Gross Margin	53.1%	51.6%	59.6%	54.4%	56.1%	59.9%
Operating Margin	42.3%	41.0%	49.6%	42.6%	45.7%	50.8%
EBITDA Margin	66.8%	67.3%	68.7%	67.0%	68.4%	68.7%
Net Margin	38.2%	37.3%	43.9%	39.4%	40.0%	44.5%
Effective Tax Rate	12.6%	10.6%	13.2%	13.1%	17.7%	17.0%

Revenue compounded at 23.2% from 1,339 billion TWD in 2020 to 3,809 billion in 2025. The trajectory is not smooth. 2023 saw a 4.5% revenue decline as the broader semiconductor cycle bottomed out, then 33.9% growth in 2024 and 31.6% in 2025 as AI demand took over.

Margins expanded substantially. Operating margin went from 42.3% to 50.8%, an 850 basis point improvement. Gross margin moved from 53.1% to 59.9%. Net margin from 37.3% to 44.5%. Q1 2026 was even stronger: net margin of 50.5%, revenue up 35% year over year, and full-year guidance raised to above 30% growth in USD terms.

3.2 DuPont Decomposition

DUPONT DECOMPOSITION						
Net Margin (NI / Revenue)	38.2%	37.3%	43.9%	39.4%	40.0%	44.5%
Asset Turnover (Revenue / Avg Assets)	0.49x	0.49x	0.52x	0.41x	0.47x	0.52x
Equity Multiplier (Avg Assets / Avg Equity)	1.50x	1.63x	1.72x	1.66x	1.59x	1.52x
ROE (NI / Avg Equity)	27.9%	29.8%	39.3%	26.9%	30.2%	35.3%

When we decompose ROE into margin, asset turnover, and equity multiplier, the story is clear: TSMC's 35% ROE comes from margin and operating efficiency, not leverage. The equity multiplier was 1.50x in 2020 and 1.52x in 2025, meaning leverage is essentially unchanged. Net margin did almost all the work.

This is a high-quality ROE profile. Companies that lever up to drive ROE create financial fragility. TSMC has done the opposite. The 2023 dip to 26.6% ROE was driven by asset turnover falling from 0.52x to 0.41x as revenue contracted, which is the textbook cyclical signature.

3.3 Risk

Liquidity and Solvency

Both metrics are about as good as they get. The current ratio is 2.51x, EBIT to interest expense exceeds 150x, and the company has more cash than debt. Default risk is irrelevant. Investment grade ratings are Aa3 (Moody's) and AA- (S&P).

Customer Concentration

This is the biggest single business risk. Apple and NVIDIA together account for more than 35% of revenue. Losing either would be a multi-quarter shock to results. The mitigation is that switching foundries at the leading edge is structurally very expensive, so the same factor that creates the risk also limits its likelihood.

Geopolitical

About 70% of TSMC's leading-edge production sits in Taiwan, which is the obvious tail risk. The Arizona, Kumamoto, and Dresden expansions are real but partial mitigations. Around 74% of the revenues come from USA, and full geographic diversification is a multi-decade project and there is no near-term scenario in which the leading edge meaningfully migrates out of Taiwan. One area to watch out, is the Taiwan and China conflict

TSM US Equity							
Filter Domestic, Sectors, or Market Cap							
Summary Key Metrics Geo-Exposure Risks Commodities/Materials Sustainability Suppliers Customers As of 12/31/2025 Currency BRL (1x0)							
Country/Region	Est Company Revenue %	Company Facilities	% Company Facilities	Suppliers Domiciled	% Suppliers Domiciled	Supplier Facilities	% Supplier Facilities
Total	---	37	---	244	---	7328	---
United States of America *	74.42%	4	10.81%	35	14.34%	1336	18.23%
Japan *	3.95%	3	8.11%	47	19.26%	1190	16.24%
China *	8.60%	2	5.41%	16	6.56%	650	8.87%
Taiwan *	7.86%	26	70.27%	97	39.75%	474	6.47%
Germany *	---	2	5.41%	5	2.05%	392	5.35%
South Korea *	---	---	---	22	9.02%	344	4.69%
France *	---	---	---	3	1.23%	252	3.44%
India *	---	---	---	---	---	243	3.32%
United Kingdom *	---	---	---	2	0.82%	176	2.40%
Italy *	---	---	---	1	0.41%	130	1.77%
Spain *	---	---	---	---	---	123	1.68%

Cyclicality

The 2023 cycle showed TSMC is not immune. Revenue fell 4.5% and operating margin compressed to 42.6%. We should expect future cycles to look similar in shape, although the AI demand pulse may dampen the amplitude relative to what we saw in 2023.

3.4 Cross-Sectional Comparison

OPERATING STATISTICS (LTM as of Q1 2026)										
Company	Gross Margin	EBITDA Margin	EBIT Margin	Net Margin	Rev Growth 1Y	EBITDA Growth 1Y	Net Income Growth 1Y	Debt/Capital	Debt/EBITDA	NTM EPS Growth
United Microelectronics	29.0%	42.4%	18.5%	17.6%	2.3%	3.4%	(11.6%)	17.2%	0.80x	13.9%
MediaTek	47.5%	21.0%	17.4%	17.7%	12.3%	2.4%	(1.0%)	2.4%	0.10x	25.9%
Broadcom	76.7%	54.5%	41.9%	36.6%	25.2%	27.7%	147.9%	45.3%	1.80x	41.2%
Novatek Microelectronics	37.7%	19.3%	18.1%	16.2%	(2.1%)	(15.6%)	(19.6%)	0.8%	-	(2.6%)
Realtek Semiconductor	50.0%	13.4%	11.7%	12.0%	8.2%	6.4%	(3.5%)	17.9%	0.70x	9.8%
SK hynix	59.6%	69.3%	58.6%	56.9%	85.0%	128.3%	189.3%	10.5%	0.20x	72.3%
ChipMOS Technologies	10.8%	25.0%	4.8%	2.1%	5.5%	2.8%	(65.1%)	40.5%	2.60x	n/m
Advanced Micro Devices	52.5%	19.5%	10.8%	12.5%	34.3%	31.0%	164.2%	6.0%	0.60x	42.6%
ASE Technology	17.7%	17.9%	7.9%	6.3%	8.4%	19.7%	25.2%	41.4%	2.20x	31.3%
ASPEED Technology	68.0%	54.5%	51.3%	43.2%	40.6%	57.6%	52.8%	4.0%	0.10x	49.6%
Taiwan Semiconductor	61.9%	69.6%	53.3%	46.5%	30.7%	32.8%	47.4%	14.6%	0.40x	26.2%

The cross-sectional comparison shows just how far ahead TSMC is. Operating margin is 50.8%, compared to 18.5% for UMC, 17.4% for MediaTek, and 7.9% for ASE Technology. Among the peer set, only Broadcom (41.9%), SK hynix (58.6%), and ASPEED (51.3%) come close, and those are very different businesses.

What's more interesting is that the gap is widening, not closing. Five years ago TSMC's operating margin was around 39% and UMC's was around 18%. The gap was 21 points. Today it's 32 points. That's not what you'd expect to see if competitive pressure was eroding TSMC's position.

4. Forecasting

We use the condensed forecast approach from class. Five years out (2026 to 2030), with revenue growth tapering linearly from 35% in Y1 to 2.5% terminal.

4.1 Assumptions

ASSUMPTIONS		
COST OF CAPITAL		
Risk-Free Rate	1.40%	Taiwan 10-year government bond yield. Currency-matched to TWD cash flows.
Equity Risk Premium	5.50%	BAV course standard mature-market premium.
Beta	1.25	TSM ADR 5-year monthly beta (Yahoo Finance). TWSE 2330 local beta is 1.15.
Cost of Equity (CAPM)	8.28%	$R_f + \text{Beta} \times \text{ERP}$
Pre-Tax Cost of Debt	3.00%	Taiwan R_f plus AA- corporate spread.
Effective Tax Rate	20.00%	Taiwan statutory corporate income tax rate.
After-Tax Cost of Debt	2.40%	$K_d \times (1 - \text{tax rate})$
WACC	8.17%	Weighted by market values. Debt is small share of capital structure.
FORECAST INPUTS		
Y1 Revenue Growth	35.00%	S&P Capital IQ analyst consensus for 2026.
Terminal Growth Rate	2.50%	Long-run nominal global GDP growth.
Operating Margin (forecast)	47.00%	Three-year historical average (2023, 2024, 2025).
Net Margin (forecast)	41.00%	Three-year historical average.
Capex as percent of Revenue	30.00%	Tapered down from 33 percent historical.
Working Capital as percent of Delta Revenue	10.00%	Standard BAV course assumption.
MULTIPLES		
Terminal P/E Multiple	22.00x	Sector multiple.
Terminal EV/EBITDA Multiple	15.00x	Sector multiple.
Industry Median ROE	15.00%	Semiconductor foundry median per Damodaran data.

Year 1 growth of 35% lines up with both Capital IQ analyst consensus and TSMC's own guidance of 'above 30%' for 2026. Operating margin holds at 47%, which is the three-year average. Net margin sits at 41%, also a three-year average and deliberately below where Q1 2026 actually came in (50.5%) and where consensus expects the full year to land. Capex tapers from the 33% historical share of revenue down to 30% over the forecast. Working capital scales at 10% of revenue change, which is the standard course assumption.

4.2 Forecasted Income Statement

5-YEAR FORECAST (TWD millions)

Line Item	Y0 (2025A)	Y1 (2026E)	Y2 (2027E)	Y3 (2028E)	Y4 (2029E)	Y5 (2030E)
Revenue Growth Rate	-	35.0%	26.9%	18.8%	10.6%	2.5%
Revenue	3,809,054	5,142,223	6,524,196	7,747,483	8,570,653	8,784,919
Operating Margin	50.8%	47.0%	47.0%	47.0%	47.0%	47.0%
EBIT	1,936,874	2,416,845	3,066,372	3,641,317	4,028,207	4,128,912
NOPAT (EBIT x (1 - tax))	-	1,933,476	2,453,098	2,913,053	3,222,565	3,303,129
Net Margin	44.5%	41.0%	41.0%	41.0%	41.0%	41.0%
Net Income	1,695,125	2,108,312	2,674,920	3,176,468	3,513,968	3,601,817

We chose 41% net margin in the base case as a floor, not a central estimate. Q1 2026 actuals and Q2 2026 guidance both point to something more like 47 to 49%. Using the lower number means our valuation is conservative, which is what you want when the market is at all-time highs. We discuss the higher-margin sensitivity later on.

4.3 Balance Sheet Roll

	(2025A)	(2026E)	(2027E)	(2028E)	(2029E)	(2030E)
Book Value Equity (end)	5,355,039	7,463,350	10,138,271	13,314,738	16,828,706	20,430,523

We assume 100% earnings retention for the AE forecast. This is a simplification because TSMC actually pays out around 27% of net income as dividends. Under clean-surplus accounting, the choice between retention and payout doesn't change the valuation. Paying dividends reduces future book value, which reduces normal earnings, which raises future abnormal earnings, exactly offsetting. The forecasted balance sheet should therefore be read as a synthetic construct.

4.4 Cash Flow

Free cash flow grows substantially over the forecast horizon as capex tapers from 33% to 30% of revenue. By Y5, FCF reaches roughly 3.2 trillion TWD. This drives a strong DCF cross-check.

5. Valuation

5.1 Cost of Capital

COST OF CAPITAL		
Risk-Free Rate		1.40%
Equity Risk Premium		5.50%
Beta		1.25
Cost of Equity (CAPM)		8.28%
Pre-Tax Cost of Debt		3.00%
Effective Tax Rate		20.00%
WACC		8.17%
Terminal Growth Rate		2.50%

We start with a Taiwan risk-free rate of 1.4% to match TWD cash flows, an equity risk premium of 5.5% (the course standard), and a beta of 1.25 (TSM ADR five-year monthly beta from Yahoo). Cost of equity comes out to 8.28%. After-tax cost of debt of 2.4% and a debt-light capital structure give a WACC of 8.17%.

One thing we want to flag: a Taiwan country risk premium of 50 to 100 basis points would be defensible given the geopolitical exposure. We use the course standard for the base case, but if we add 75 bps to K_e our AE valuation drops by roughly 200 TWD per share.

5.2 Abnormal Earnings

AE values equity as current book value plus the present value of future abnormal earnings, where abnormal earnings are net income less the normal return on book equity (K_e times beginning book value).

ABNORMAL EARNINGS VALUATION					
Component	Y1	Y2	Y3	Y4	Y5
Beginning Book Value	5,355,039	7,463,350	10,138,271	13,314,738	16,828,706
Net Income	2,108,312	2,674,920	3,176,468	3,513,968	3,601,817
Normal Earnings ($K_e \times \text{Beg BV}$)	443,129	617,592	838,942	1,101,795	1,392,575
Abnormal Earnings	1,665,182	2,057,328	2,337,526	2,412,173	2,209,241
Discount Factor (K_e)	0.9236	0.8530	0.7878	0.7276	0.6720
PV of Abnormal Earnings	1,537,919	1,754,879	1,841,501	1,755,075	1,484,575

Current Book Value (Y0)	5,355,039
Sum PV of AE (Y1 to Y5)	8,373,949
Terminal AE Y6 ($Y5 \text{ AE} \times (1 + g)$)	2,264,472
Terminal Value (Gordon, end of Y5)	39,211,642
PV of Terminal AE	26,349,603
AE Equity Value	40,078,590
AE Value Per Share (TWD)	1,545.50

Our AE result is 1,545.5 TWD per share, 29% below the market price of 2,185. Across the five-year forecast horizon, abnormal earnings remain firmly positive in every year, rising from TWD 1.67M in Y1 to a peak of TWD 2.41M in Y4 before tapering to TWD 2.21M in Y5. This profile tells a consistent story, that the company is expected to earn substantially more than its cost of equity throughout the explicit forecast period, confirming that it is a genuine value creator rather than merely a value-preserving business.

5.3 Sensitivity

Our base case uses inputs that we know are conservative. If we use 47% net margin (consistent with consensus and Q1 2026 actual), the AE valuation rises noticeably. At 49% (the midpoint of Q2 2026 guidance) it rises further. The base 41% is meant as a floor, not a midpoint estimate, and we want to be transparent about that.

Year 1 growth has surprisingly little impact on AE. The reason is that revenue growth flows through to net income via a fixed margin, and book value grows in lockstep, so the spread between net income and normal earnings doesn't widen sharply. Margin assumptions matter much more than growth rate assumptions in this model.

5.4 ROE Mean Reversion

VALUATION IMPACT

Scenario	PV of AE Y1 to Y10	PV of Terminal BV	Total Equity Value	Per Share (TWD)	vs Current Price
1. Sustain ROE	43,553,817	5,355,039	48,908,856	1,886.01	(13.7%)
2. Fade to Industry (10Y)	15,263,238	5,355,039	20,618,276	795.07	(63.6%)
3. Fade to Cost of Equity (10Y)	9,730,419	5,355,039	15,085,458	581.72	(73.4%)
4. Fast Fade (5Y)	8,282,677	5,355,039	13,637,716	525.89	(75.9%)

This is the most useful piece of analysis we did. The question is: how persistent is TSMC's abnormal ROE? Right now ROE is 35.3% and K_e is 8.28%, so there are 27 percentage points of abnormal returns. Where do they go from here? We ran four scenarios.

In Sustain, ROE holds at 35.3% for ten years. The implied value is 1,886 TWD per share, still 13.7% below market. In Fade to Industry, ROE drops linearly to 15% over ten years, giving 795 TWD. In Fade to Cost of Equity, ROE drops to 8.28%, giving 582 TWD. In Fast Fade (5-year drop to industry), 526 TWD.

What's striking is that even Sustain doesn't justify the current price. The market is implicitly pricing either further ROE expansion above 35% or a much longer high-ROE plateau than ten years. Both are possible (AI cycle durability is the main bull case) but neither falls naturally out of the model.

5.5 DCF Cross-Check

DCF VALUATION (CROSS-CHECK)					
Component	Y1	Y2	Y3	Y4	Y5
Free Cash Flow	1,800,159	2,314,900	2,790,725	3,140,248	3,281,703
Discount Factor (WACC)	0.9245	0.8546	0.7901	0.7304	0.6753
PV of FCF	1,664,194	1,978,421	2,204,939	2,293,700	2,215,976
Sum PV of FCF	10,357,230				
Terminal FCF Value (undiscounted)	59,325,316				
PV of Terminal Value	40,059,536				
Enterprise Value	50,416,766				
Less: Net Debt (TWD mm)	(2,000,078)				
DCF Equity Value	52,416,844				
DCF Value Per Share (TWD)	2,021.28				

DCF gives us 2,021 TWD per share, only 7.5% below market. The DCF result is meaningfully higher than AE because (a) FCF benefits from capex tapering in the forecast and (b) net cash gets added back explicitly to enterprise value. The two methods bracket what we think is a reasonable intrinsic value range.

5.6 Sensitivity Grid

WACC vs Terminal Growth (DCF per-share value, TWD)					
WACC (down) / g (right)	1.00%	1.75%	2.50%	3.25%	4.00%
6.17%	2,305.76	2,628.98	3,084.44	3,774.17	4,941.49
7.17%	1,930.44	2,143.12	2,424.16	2,812.85	3,385.68
8.17%	1,660.25	1,809.09	1,997.34	2,243.02	2,577.16
9.17%	1,456.54	1,565.46	1,698.89	1,866.15	2,081.97
10.17%	1,297.55	1,379.99	1,478.57	1,598.53	1,747.66

The grid shows how DCF per share moves with WACC and terminal growth. To justify the current price the market would need to be using a WACC closer to 7.5% with our terminal growth, or terminal growth closer to 3.25% with our WACC.

5.7 Multiples

TRADING MULTIPLES (LTM and NTM)

Company (Ticker)	TEV/Rev (LTM)	TEV/EBITDA (LTM)	TEV/EBIT (LTM)	P/E (LTM)	P/Tang BV (LTM)	TEV/Rev (NTM)	TEV/EBITDA (NTM)	P/E (NTM)
United Microelectronics (TWSE:2303)	3.60x	8.30x	18.70x	22.00x	2.40x	3.31x	7.39x	18.72x
MediaTek (TWSE:2454)	6.10x	28.90x	35.10x	36.90x	12.10x	5.83x	30.24x	38.51x
Broadcom (NASDAQ:AVGO)	30.10x	54.90x	71.80x	82.40x	n/m	17.07x	24.92x	31.39x
Novatek Microelectronics (TWSE:3034)	2.10x	10.80x	11.60x	15.50x	3.90x	2.10x	11.17x	15.71x
Realtek Semiconductor (TWSE:2379)	1.90x	14.20x	16.40x	19.80x	5.90x	1.73x	11.80x	17.03x
SK hynix (KOSE:A000660)	6.60x	9.50x	11.30x	12.10x	5.70x	2.24x	2.64x	3.98x
ChipMOS Technologies (TWSE:8150)	2.00x	8.00x	48.80x	98.30x	2.00x	1.69x	6.10x	18.56x
Advanced Micro Devices (NASDAQ:AMD)	16.20x	79.30x	148.80x	133.30x	26.80x	11.98x	48.72x	51.82x
ASE Technology (TWSE:3711)	3.70x	20.00x	45.70x	55.70x	7.70x	3.13x	13.99x	33.26x
ASPEED Technology (TPEX:5274)	65.90x	120.20x	128.50x	154.20x	86.90x	40.68x	72.60x	91.62x
Taiwan Semiconductor (TWSE:2330)	13.70x	19.70x	25.70x	30.80x	10.00x	10.19x	13.95x	21.72x

Summary Statistics (peers, excludes TSMC)

High	65.90x	120.20x	148.80x	154.20x	86.90x	40.68x	72.60x	91.62x
Low	1.90x	8.00x	11.30x	12.10x	2.00x	1.69x	2.64x	3.98x
Mean	13.82x	35.41x	53.67x	63.02x	17.04x	8.98x	22.96x	32.06x
Median	4.90x	17.10x	40.40x	46.30x	5.90x	3.22x	12.90x	25.06x

Forward multiples produce a wide range. P/E of 22x on Y1 EPS of 81 TWD gives 1,789 per share. EV/EBITDA of 15x on Y1 EBITDA of 3.93 trillion gives 2,367. The peer set median NTM P/E is 25.1x, and TSMC trades at 21.7x, suggesting it's not aggressively priced relative to peers.

5.8 Putting it Together

VALUATION SUMMARY		
Method	Per-Share Value (TWD)	vs Current Price
Abnormal Earnings (primary)	1,545.50	(29.3%)
DCF	2,021.28	(7.5%)
P/E Multiple (22x)	1,788.60	(18.1%)
EV/EBITDA Multiple (15x)	2,367.41	8.3%
Median (blended target)	1,904.94	(12.8%)
Current Price (TWD)	2,185.00	

Our methods give a per-share range of 1,545 to 2,367 TWD, with median 1,905. The current price of 2,185 is at the upper end of that range but below the multiples ceiling. AE is the lowest figure because it's the most conservative method.

The price-value gap is best explained by expectations rather than risk. If we hold our cost of capital and beta fixed, the market price implies a long-run growth rate of 4.4%, well above our 2.5% terminal.

6. Recommendation

6.1 Summary

RECOMMENDATION	
FINAL RECOMMENDATION	
Current Market Price (TWD)	2,185.00
Blended Target Price (median, TWD)	1,904.94
Implied Upside / (Downside)	(12.8%)
Recommendation	HOLD

VALUATION BY METHOD		
Method	Per-Share Value (TWD)	vs Current Price
Abnormal Earnings (primary)	1,545.50	(29.3%)
DCF	2,021.28	(7.5%)
P/E Multiple (22x)	1,788.60	(18.1%)
EV/EBITDA Multiple (15x)	2,367.41	8.3%

Our intrinsic value range is 1,545 to 2,367 TWD per share with a median of 1,905. The market price of 2,185 sits 13% above our median. Even our most aggressive scenario (Sustain ROE at 35% for ten years) only gets us to 1,886, still below market. The recommendation is HOLD.

6.2 Why the Gap Exists

AI cycle durability

This is the most credible explanation for why the market is priced where it is. Q1 2026 net margin was 50.5%, which is well above our 41% base case and even above the 47 to 49% sensitivity range. Management raised long-term gross margin guidance to 56% or more on the April 2026 earnings call. If those margin levels stick, the AE valuation rises substantially.

Whether they stick depends on how durable the AI capex cycle is. If hyperscaler training capex stays aggressive through 2028 to 2030, TSMC's pricing power and margins probably persist. If AI capex normalizes earlier, margins compress as utilization falls.

Higher long-term growth

The market is implicitly pricing about 4.4% long-run growth versus our 2.5%. This could be defensible if TSMC's mix continues to shift toward leading-edge nodes where pricing power is highest. We just don't think it should be the base case for a company already generating 3.8 trillion TWD of revenue.

Underpriced geopolitical risk

About 70% of TSMC's leading-edge capacity is in Taiwan. If we add a 75 basis point country risk premium to Ke, our AE valuation falls by roughly 200 TWD. The market does not appear to be pricing in this scenario at all. That's an asymmetric tail risk that gets ignored in normal times and becomes the entire story when something happens.

Momentum

TSMC hit an all-time high of 2,190 TWD on April 24, 2026, the day before our cutoff. The 52-week range is 893 to 2,190. The stock is up 41% year to date. Some of the price-value gap is probably just momentum and sentiment rather than fundamentals.

6.3 Why HOLD and not SELL

The mechanical recommendation from the median target is technically a SELL (downside between 10 and 25%), but we are softer than that for a few reasons. The Sustain scenario at 1,886 implies only 13.7% downside, which is within normal single-stock volatility. Q1 2026 was a clear positive surprise and management raised full-year guidance. TSMC is unusually durable: switching costs at the leading edge are very high and Samsung Foundry and Intel Foundry have not yet caught up. The dividend yield of about 1% provides a small carry. And sell-side consensus is 2,509 TWD, implying 14.8% upside. We're below consensus but not aggressively so. Forcing a SELL recommendation here would require us to be more confident than the data supports.

6.4 Why HOLD and not BUY

Every method we ran produces an intrinsic value at or below the current price under base-case assumptions. The stock is at an all-time high after a 41% year-to-date run. The implied long-term growth in the price is above the long-run nominal GDP rate. The geopolitical tail risk isn't being priced in. And semiconductor cycles do eventually turn, as 2023 reminded us. Buying at fresh all-time highs is rarely the right call when the model tells you the stock is expensive.

6.5 What to Watch

If we were holding the stock, these are the things we'd be tracking:

- Quarterly margin progression. If TSMC continues to print net margin above 48% through 2026, our higher-margin sensitivity becomes the central case and the gap to market price closes.
- AI capex commentary from NVIDIA, Broadcom, AMD, and the hyperscalers. Their forward guidance drives TSMC's leading-edge utilization.
- Foundry market share. If Samsung Foundry or Intel Foundry win meaningful AI accelerator volumes, TSMC's pricing power compresses.
- Geopolitical developments. Any material Taiwan-China escalation would dramatically increase our country risk premium.
- TWD/USD exchange rate. TSMC reports in TWD but earns about 74% of revenue from US customers. A 5% TWD appreciation cuts roughly 5% from reported revenue and earnings growth.

6.6 Final Note

TSMC is one of the most exceptional businesses in modern equity markets. A 35% ROE, 50% operating margins, 2 trillion TWD of net cash, and 70% of the global foundry market put it in a very small group of structurally dominant companies. Our HOLD recommendation isn't a comment on the quality of the business. It's a comment on the price relative to intrinsic value at this moment in time.